

ASSESSING THE RELATION BETWEEN ESG PERFORMANCE AND FINANCIAL RETURNS

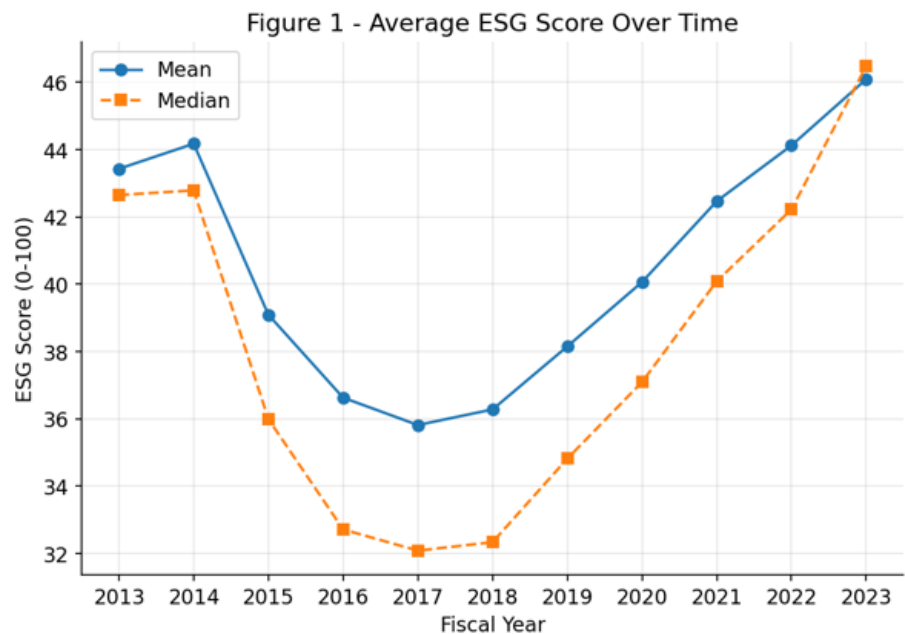
RSM8224

Analytical Insights Using
Financial and Accounting Data

Presented by Group 9

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Part I: Distribution of ESG Score



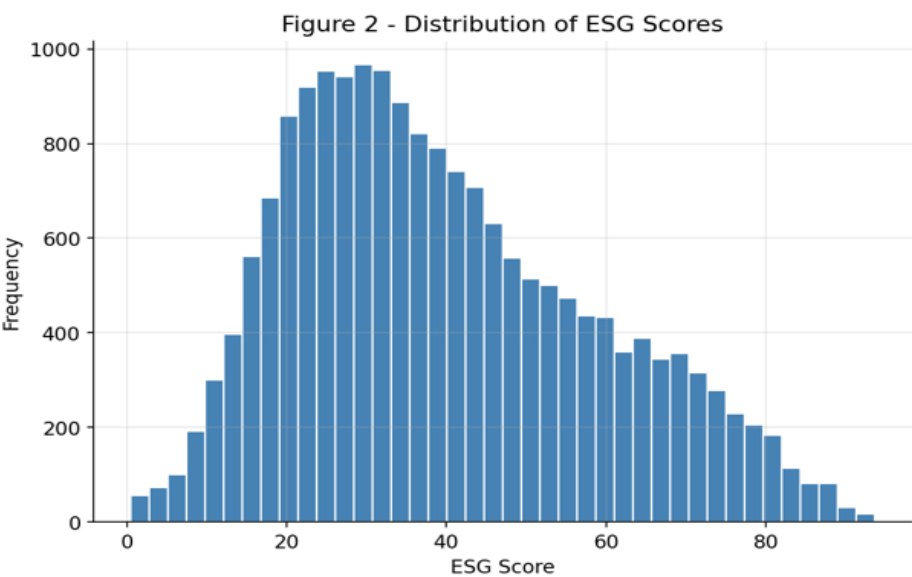
ESG Distribution across Time

Pre-2017 Decline

- Downward trend primarily reflects **dataset broadening**
- Entry of **newly reporting firms** with structurally lower ESG baselines

Post-2017 Increase

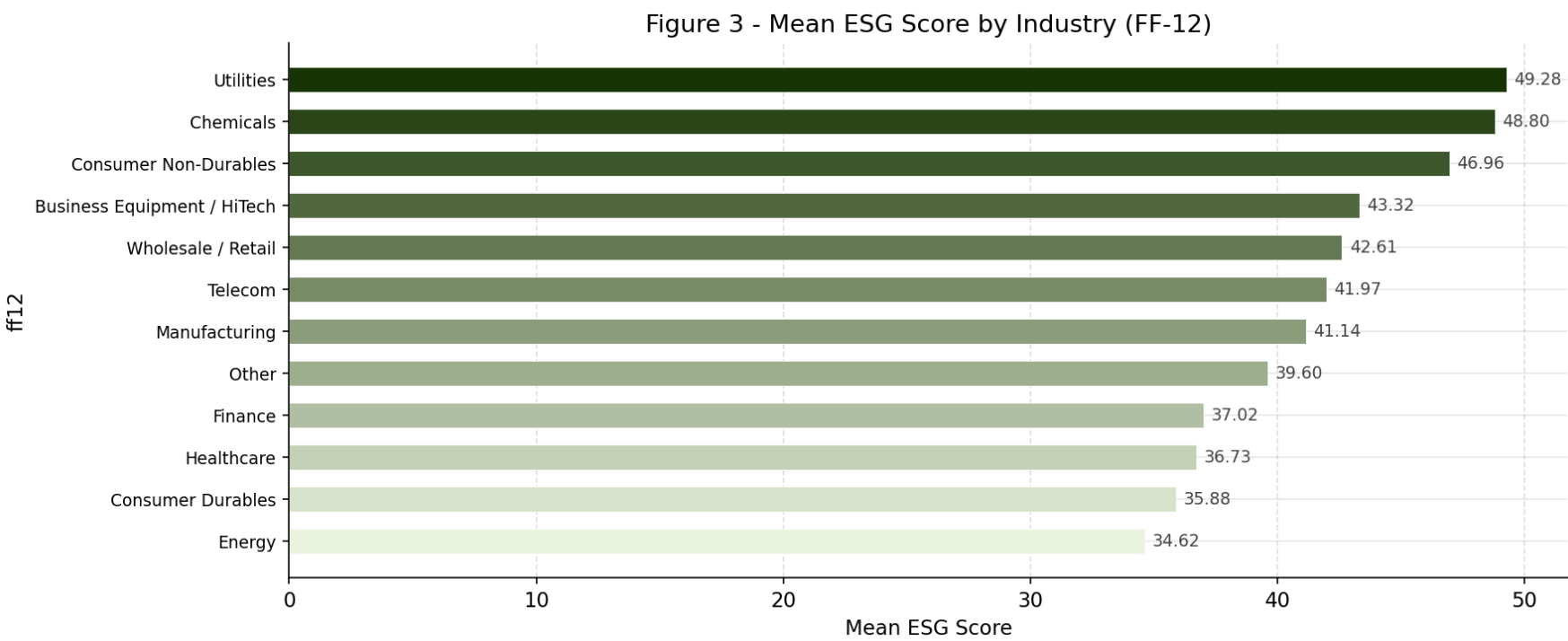
- Clear inflection point with sustained score improvement
- Driven by rising **regulatory pressure** and **stakeholder focus** on ESG



ESG Distribution across Firms

- **Weak performance in general around 35**
- **Fat tail of high-performers above 60**
- **Wide difference across firms**

Part I: ESG Score across Industries



ESG Leaders

- **Utilities:** Regulation-driven disclosure & Consumer-Facing
- **Chemicals:** Heavy compliance & Large-firm advantage
- **Consumer Non-Durables:** Reputation & consumer pressure

ESG Laggards

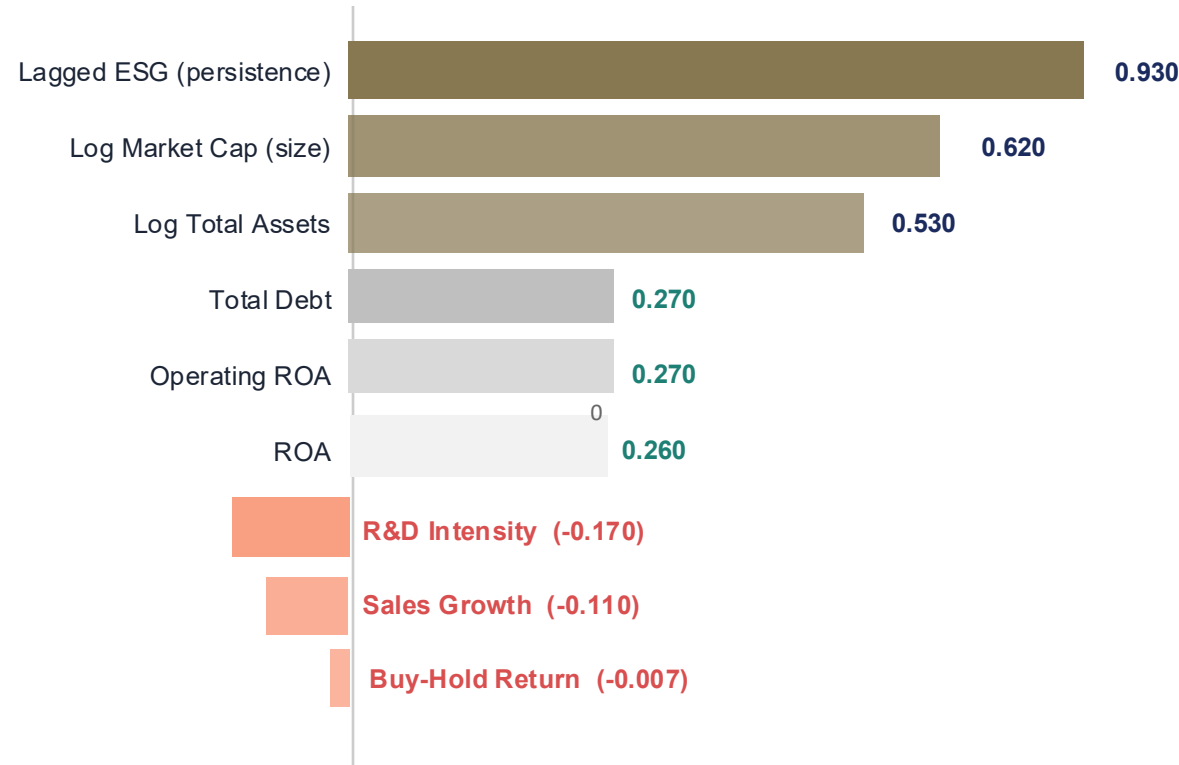
- **Energy:** High emissions penalized & Disclosure resistance
- **Consumer Durables:** Fragmented with under-resourced reporting
- **Healthcare:** R&D focus with disclosure gap
- **Finance:** Profitability focus with disclosure gap

Part I: Key Insights from Cross-Industry Comparison



Part II: Key Firm Characteristics That Determines ESG Score

Correlations with ESG Score



Key Insights:

- ESG is highly persistent
- Size dominates
- Stock returns show near zero raw correlation

Multivariate regression

Variable	M1: No FE	M2: +Year & Industry FE
Firm size	+6.68***	+7.11 ***
Profitability	+0.70	-1.72 *
Leverage	+5.91***	+2.87 ***
R&D intensity	+0.16	-11.00 ***
Book-to-mkt	+4.53***	+4.40 ***
Sales growth	-1.58***	-1.94 ***
R ²	0.387	0.437

Firm size is the only robust and stable driver of ESG

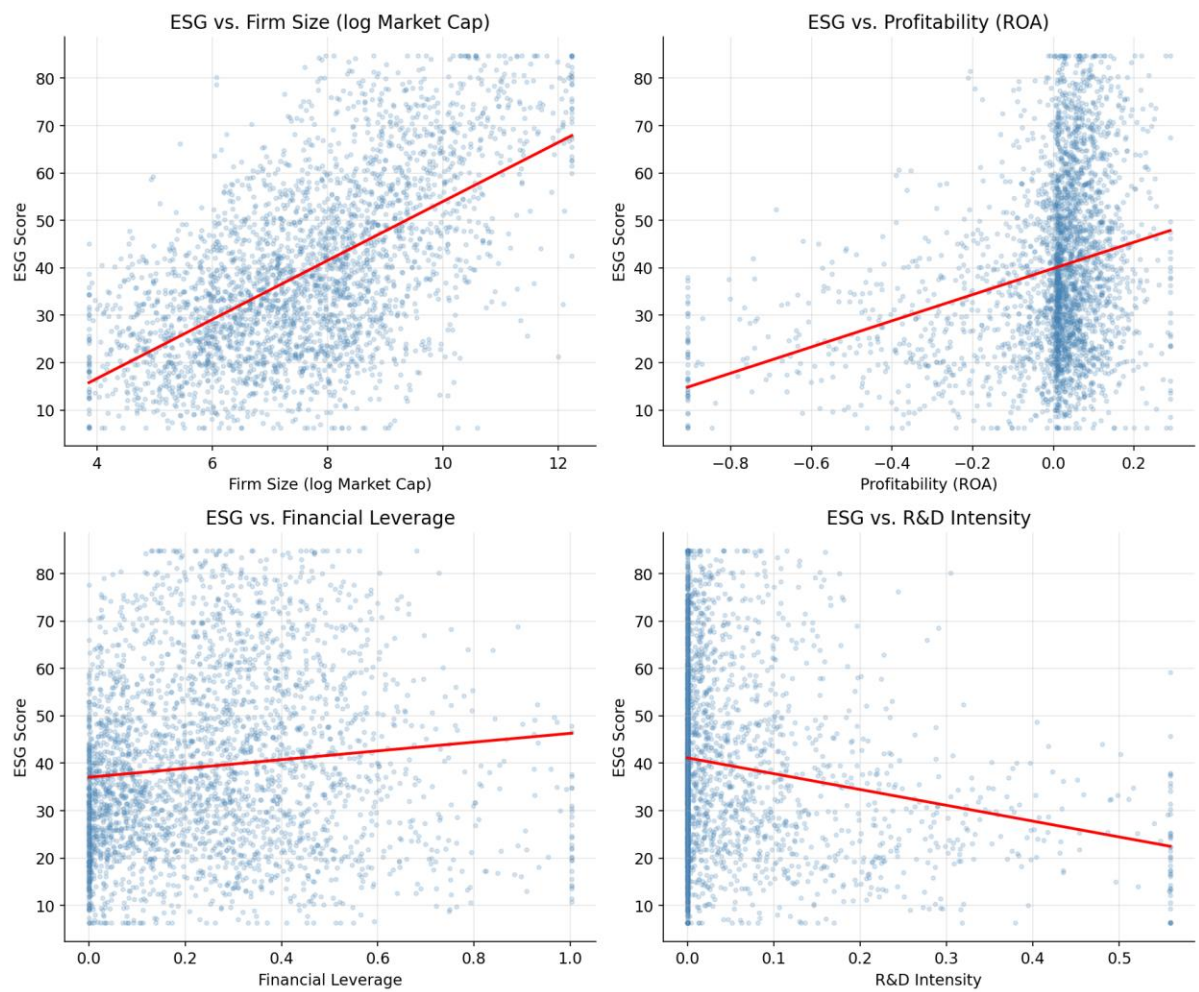
Strategic Tradeoff:

- Within the same industry, R&D-intensive firms clearly spend less on ESG
- Value firms tend to score higher
- High-growth firms invest less in ESG

Short term profitability is not a true determinant

Part II:4 Key Firm Characteristics

Scatter Plot of the Four characteristics with ESG Score



Quartile Analysis

	Q1	Q2	Q3	Q4	
Size	26.9	33.9	40.1	56.9	Monotonic ↑
Profitability	31.5	37.8	43.8	44.6	Threshold Effect
Leverage	32.0	39.6	44.4	41.6	Hump-shaped
R&D intensity	41.7	36.8	43.7	35.5	Non - monotonic

Variable	Direction	Reasoning
Firm size	↑	Scrutiny, resources, capacity
Profitability	↑	Slack to fund ESG, up to a threshold
Leverage	↑	Moderate debt helps; extreme debt constrains
R&D intensity	↓	Competes with ESG for same budget

Part III: Model Setup – Two Channels of Financial Materiality

Research Question

Does ESG have financial materiality through the accounting channel (ROA) and the market-pricing channel (stock returns)?

Model Specification (R1 → R4)

R1	ESG only (unconditional)
R2	+ controls + year FE + industry FE
R3	R2 + lagged ROA (earnings persistence)
R4	Firm FE + year FE (within-firm test)

Model Specification (M1 → M7)

M1	ESG only (unconditional)
M2	+ firm controls + year FE + industry FE
M3	M2 + earnings news (Δ ROA)
M4	M2 + current ROA level (instead of Δ ROA)
M5	M3 + prior 12-month momentum
M6	Firm FE + year FE
M7	Firm FE + year FE + momentum

Variables & Specification -- Shared across both channels

Dependent variable
roa_w — winsorized ROA
bhr_w — winsorized future 12-month buy-and-hold return.

Main explanatory variable
esg_lag — lagged Refinitiv ESG score (0–100).

Firm controls (aligned with Part III-A)
log_me, bm_w, leverage_w, sales_growth_w, rd_intensity_w.

Earnings-news proxy
earnings_news_w = Δ ROA. Accounting-based proxy for unexpected earnings (no IBES forecasts in the panel, so not a standard SUE).

Fixed effects
Year FE in M2–M7; industry FE (FF-12) in M2–M5; firm FE + year FE in M6–M7.

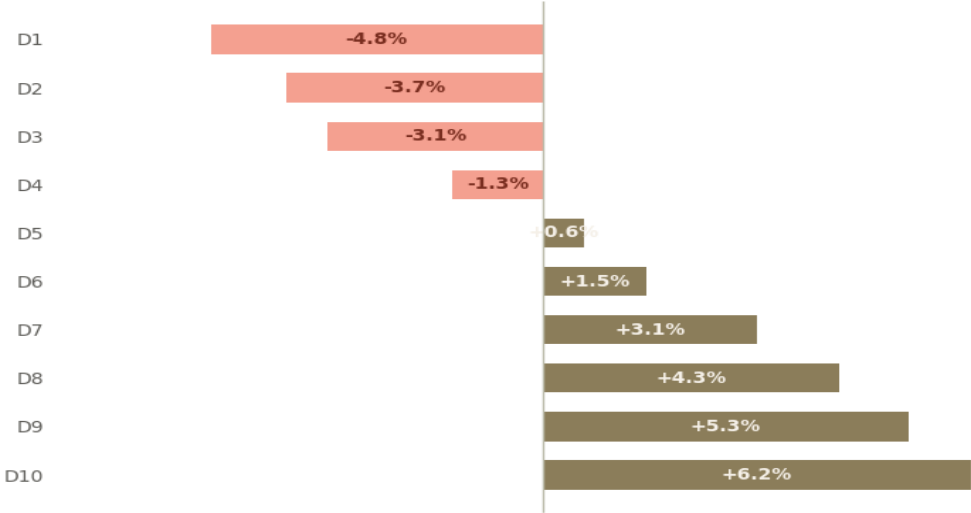
Part III-A: Does ESG predict future ROA?

Table — Regression Result

	R1	R2	R3	R4
Lagged ESG	0.0019***	-0.001	-0.0001*	-0.0001
Lagged ROA	----	----	0.545*	0.028
Controls	X	✓	✓	✓
Year / Ind FE	X	✓	✓	Time FE
Firm FE	X	X	X	✓
Adj. R ²	0.054	0.509	0.662	0.251

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Standard errors clustered at the firm level (gvkey). Controls and FE mirror Part III-A setup.

Mean ROA by Lagged ESG Decile



Raw monotonic relation — but driven by firm size, not ESG

Key Takeaways

- Raw relation is strong** — but a classic omitted variable story: ESG proxies for firm size ($r = 0.62$), which independently predicts profitability
- ESG collapses to zero after controls** — R2 Adj. R^2 jumps from 5% $\rightarrow$ 51%, showing firm characteristics drive profitability, not ESG
- No operations channel** — ESG does not independently improve ROA; financial materiality via accounting performance is not supported

Part III-B: Regression Results — ESG → Future 12-Month Return

Method: Lagged ESG coefficient across 7 specifications

1. No unconditional effect

- M1 ESG coefficient is insignificant ($p=0.431$) — ESG alone does not predict returns.

2. Positive after controls + FE

- M2–M4 coefficient stabilises at 0.0008 ($p \leq 0.003$) — ESG carries incremental pricing info.

3. Robust to earnings & momentum

- Survives Δ ROA, ROA level, and prior 12-m return (M5: 0.0007, $p=0.012$).

4. Firm FE weakens the result

- M6–M7 insignificant — evidence is cross-sectional, not within-firm over time.

5. Interpretation

- Markets price ESG as a firm-quality signal, but ESG changes within a firm don't move its own stock much.

Economic Magnitude

+1 SD in lagged ESG (≈ 19 pts) $\times$ M3 coefficient (**0.0008**) $\approx$ **+1.5 pp** higher future 12-month BHR

Part III-B: Interpretation & Link to Part III-A

Two channels of financial materiality

Part III-A — Accounting Channel

What it captures: The *fundamental* side — whether ESG translates into better realized earnings.

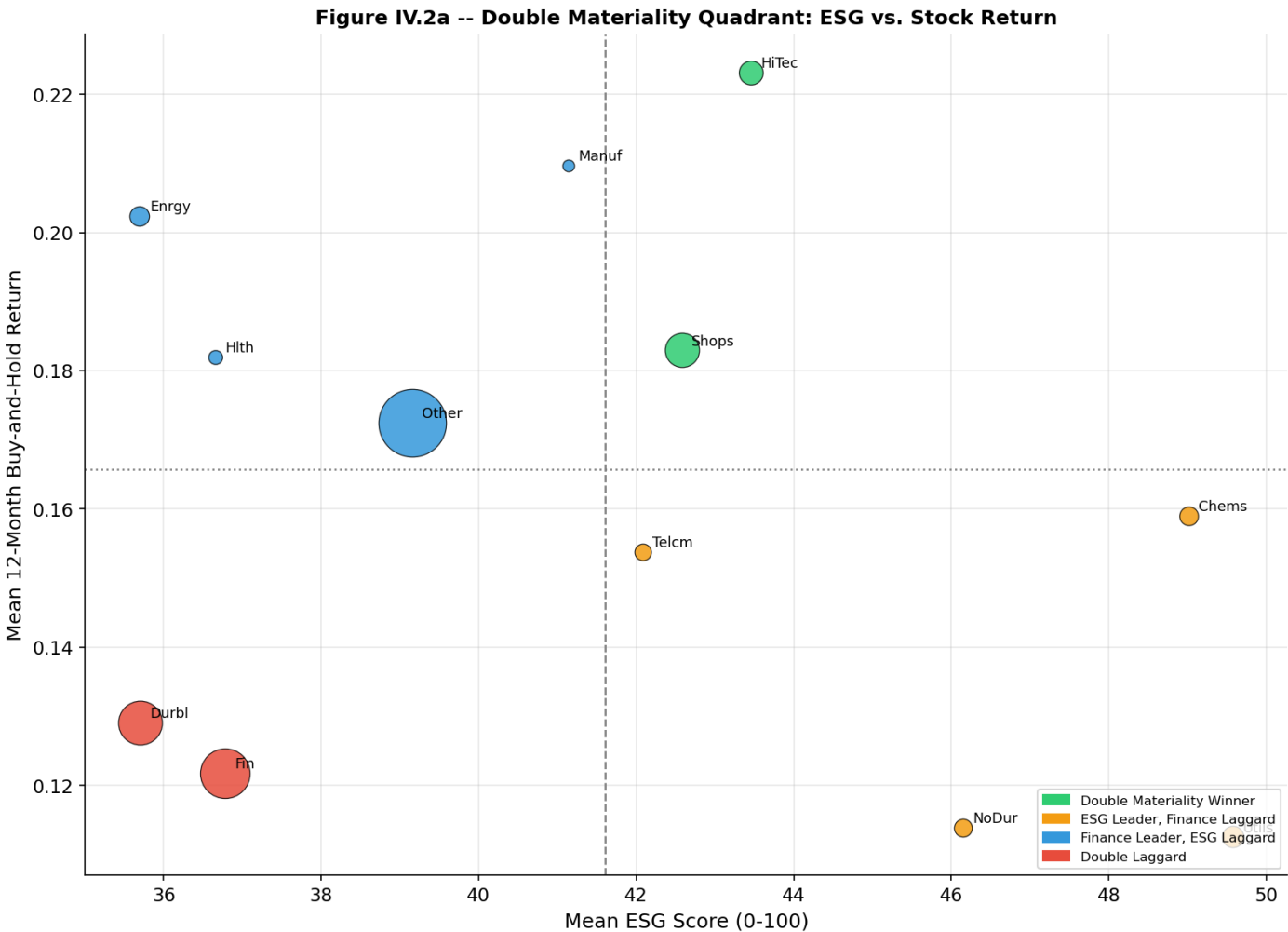
Part III-B — Market-Pricing Channel

What it captures: The *pricing* side — whether markets have priced ESG in, or are still catching up.

Synthesis — what the two parts say together

ESG serves as a market "quality label" for investors that drives a 1.5 pp return premium, despite having no measurable impact on internal firm performance or ROA.

Part IV: Double Materiality Quadrant



We split each FF12 industry into four groups using the sample median of mean ESG and mean 12-month buy-and-hold return.

Green (“Double materiality winners”): HiTec and Shops, above-median ESG and above-median returns.

Blue vs. orange show tension: strong returns with weaker ESG (Manufacturing, Energy, Healthcare, Other) or stronger ESG with weaker returns (Chemicals, Telecom, Non-durables).

Red (“double laggards”): Finance and Durables, below median on both.

This is descriptive and motivates how we label “winners” versus ESG-finance trade-offs

Part IV: Rationality of Double Materiality Winners

BHR Based: HiTech

Talent war drives ESG

- Tech firms compete fiercely for top talent
- They invest heavily in employee benefits, diversity, and strong governance
- These happen to be core ESG metrics, but the motivation is hiring, not sustainability

Sector tailwind, not ESG:

- The 2013-2023 period was the FAANG era, tech stocks surged market-wide regardless of ESG performance;
- High BHR reflects a macro sector boom, not a reward for doing good

Correlation, not causation

- High ESG and high BHR are driven by two separate forces that happen to coexist in the same industry
- The kind of "false winner" investors should scrutinize carefully

ROA Based: Shops

Brand reputation is profit protection

- Retailers face consumers directly, a supply chain scandal or product safety failure immediately destroys sales
- Managing ESG well is really just protecting the bottom line

Supply chain excellence counts as ESG

- A retailer's core competency is running a massive, efficient supply chain
- Doing it well reduces costs and boosts ROA, while simultaneously scoring high on ESG's environmental and social criteria

Both stem from the same root capability:

- Retailers with strong operations naturally score high on both ESG and ROA
- Two expressions of the same underlying management quality, not one causing the other

Part IV: Investment Implications for ESG-Conscious Investors

Use industry tilts as a starting filter, not the whole decision

If you care about both ESG scores and past average stock returns in our sample, overweighting HiTec and Shops (and your notebook's HiTec focus) is a simple way to narrow the universe, then still pick stocks, check valuation, risk, and your mandate inside those industries.

Match the portfolio to what “success” means for you

If your goal is “I still want solid profitability, not just a hot stock market decade,” put more weight on industries that look strong on ESG and ROA together in our numbers, Shops, Chems, NoDur, Telcm, instead of assuming the same names that win on returns will win on profits.

Don't treat “raising ESG” as a timing signal

Our firm-level results suggest the return link is much clearer between different firms than when one company slowly improves its own ESG over time. Practical takeaway: use ESG more like a cross-sectional screen or risk bucket (who you're willing to own), not as a rule that “if we improve ESG next year, we'll beat the market.”

Consistent with Parts II & III: ESG functions as a firm quality signal, financially material where ESG practices directly drive product market outcomes.

Thank you for listening

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Appendix: Where ESG and Finance Diverge

High ESG, Low Returns

- **Utilities:** Regulated disclosure, high ESG but thin margins
- **Over-investment:** ESG spend without proportional financial payoff
- **Materiality type:** ESG activities generate impact materiality, not financial materiality

Low ESG, High Returns

- **HiTech & Energy:** Return-driven; ESG reporting still maturing
- **Unpriced risk:** Higher returns may reflect unpriced transition risk
- **Bolton & Kacperczyk (2021):** Carbon risk commands a return premium

Implication: ESG-return relationship is heterogeneous, industry context matters for both impact and financial materiality

Sample & Data: Sample Computation

Sample Attrition Funnel

Step	Description	Observations
Start	Compustat annual (2012-2023, at>0, INDL/STD/C/D)	101,009 obs
After CCM Link	Matched to CRSP permno (LC/LU/LS, C/P, date window)	60,177 obs
After CRSP return filter	Compute 12-month returns (at least 12 month BHR window)	19,193 obs
Final Sample	Non-missing at, ESG, BHR, ME (fyear 2013-2023)	18,443 obs 3,086 firms

Key Sample Facts

- **Period**
Fiscal Year 2013-2023, U.S. domestic common stocks
- **Data Sources:** Three datasets merged
Refinitiv ESG (tr_esg) + Compustat (comp.funda) + CRSP (crsp_msf), linked via CUSIP8 and CCM
- **Return Window**
12-month buy-and-hold starting 4 months after fiscal year-end to avoid look-ahead bias on earnings
- **ESG Score**
Base ESG score (item 1, 0-100 Scale), Not controversy-adjusted, following conventional research literature

- Grow steadily from 739 in 2013 | Peak at round 2400 in 2020-21: Growth in dataset coverage
- Dropping to 236 in 2023 due to fiscal year return data out-of-reach

Descriptive Statistics

Analysis Variables Created

► Profitability	roa & operating roa
► Size	log of market equity & log of total assets
► Valuation (BM)	book-to-market ratio
► Leverage (LEV)	total debt scaled by total assets
► Growth	year-over-year sales growth rate
► R&D	R&D expenditure scaled by total assets
► Lagged Variables	Lagged ESG & Lagged ROA
► Returns (BHR)	Buy-and-hold return (4+12 month after fiscal year end)
► Industry	Fama-French 12-industry group

Key Descriptive Statistics After Winsorization (1/99)

	ESG	ROA	BHR	ln(ME)	LEV	BM
N	18443	18442	18443	18443	18443	18443
Min	0.459	-92.3%	-83.1%	0.872	0.00%	-0.427
Mean	39.7	-0.97%	15.5%	7.72	26.1%	0.509
Median	36.4	2.44%	5.28%	7.66	22.7%	0.386
Max	93.6	29.4%	330%	14.9	99.7%	2.94
Stdev	19.0	18.1%	62.9%	1.84	22.1%	0.510

- Missing Data: Only Growth and Lagged Variables have 12.8% missing data due to inavailable prior-year data.
- Variables are **winsorized** at 1st and 99th percentile to prevent outlier & record error influence.

	M1	M2	M3	M4	M5	M6	M7
Lagged ESG	−0.0002	0.0008***	0.0008***	0.0008***	0.0007**	0.0007	0.0006
	<i>(p=0.431)</i>	<i>(p=0.002)</i>	<i>(p=0.003)</i>	<i>(p=0.002)</i>	<i>(p=0.012)</i>	<i>(p=0.265)</i>	<i>(p=0.384)</i>
Firm controls	No	Yes	Yes	Yes	Yes	Yes	Yes
Earnings news (Δ ROA)	No	No	Yes	No	Yes	No	No
ROA level	No	No	No	Yes	No	No	No
Momentum	No	No	No	No	Yes	No	Yes
Year FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Industry FE (FF-12)	No	Yes	Yes	Yes	Yes	—	—
Firm FE	No	No	No	No	No	Yes	Yes
N (firm-years)	15,737	15,737	15,737	15,737	14,983	15,737	14,983

Appendix: Industry Snapshot

Industry	ESG Score (Mean)	BHR (Mean)	ROA (Mean)
Utilities	49.6	0.11	0.01
Chemicals	49.0	0.15	0.03
Consumer Non-Durable	46.1	0.11	0.06
Manufacturing	43.5	0.22	0.01
HiTech	42.6	0.18	0.05
Telecom	42.1	0.15	0.04
Shops	41.1	0.20	0.03
Finance	39.2	0.17	0.01
Healthcare	36.8	0.12	0.02
Energy	36.7	0.18	-0.00
Other	35.7	0.12	-0.17
Consumer Dur	35.7	0.20	-0.03

This table ranks Fama-French 12 industries by average Refinitiv ESG and reports mean buy-and-hold return (BHR) and mean ROA in our sample.

Utilities and Chemicals sit at the top on ESG, but their returns and ROA are only moderate, so “high ESG” does not automatically mean “best stock performance.”

Consumer Non-durables stands out with strong ESG and the highest ROA, a pattern consistent with “doing good and doing well” at the industry level. Manufacturing and Shops show strong market returns with mid-range ESG, highlighting heterogeneity.

Other combines weak ESG and very weak ROA; Energy and Consumer durables sit low on ESG with weak or negative profitability.

These are descriptive averages, they motivate the quadrant labels and our recommendation, not causal evidence.